Domain Adaptation for Object Detection Based on Domain-Aware Prompting in Remote Sensing Imagery

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***Abstract*—Domain adaptation (DA) is of critical importance in practical remote sensing object detection, aiming to address the performance degradation caused by significant domain discrepancies between a labeled source domain and an unlabeled target domain. The DA method enables the alignment of cross-domain features, thus improving detection performance in the target domain while avoiding the high cost of manual annotation. However, most existing methods focus on mitigating domain bias by optimizing discriminative visual encoders. These coarse feature alignment strategies often overlook domain-aware semantic consistency, which leads to feature misalignment, especially for the remote sensing scene with small objects and diverse backgrounds. To address this problem, we propose a novel domain-aware prompting (DAP) framework to learn domain-invariant semantics by modeling the learnable domain-aware prompts. Specifically, we first design a reciprocal prompt interaction learning (RPIL) module to generate domain-aware semantics for each domain by modeling reciprocal interactions between textual and image modalities. Subsequently, we develop a PDPA module to capture shared domain prompt semantics from both local and global levels by constructing domain prompt prototypes. Extensive experiments demonstrate the efficacy of our proposed approach, achieving superior performance on challenging remote sensing datasets. The code is publicly accessible at https://github.com/XZhang878/DAP/**

***Index Terms*—Domain adaptation (DA), domain-aware prompting (DAP), prototype-driven domain prompt alignment (PDPA), reciprocal prompt interaction learning (RPIL), remote sensing object detection.**

I. Introduction

DEEP learning-based object detection has witnessed remarkable advancements over the last decade, becoming a cornerstone in remote sensing image (RSI) analysis. However, a critical limitation persists: mainstream methodologies heavily rely on data-driven, supervised architectures [1], [2], [3]. These approaches necessitate extensive and often prohibitively expensive labeled datasets. Furthermore, the inherent variability of remote sensing data poses a significant challenge for model generalization. RSIs are collected under diverse conditions, including differences in geographic regions, weather conditions, ground sampling distances, and viewing angles. As a result, discrepancies in data distributions between datasets are inevitable, leading to the well-known issue of domain shift [4], [5], [6], [7], [8]. This sensitivity to distributional shifts underscores the need for more robust and data-efficient object detection techniques in remote sensing applications.

To address the issue of domain shift, many domain adaptation (DA) object detection techniques have been proposed, which leverage labeled source domain to enhance detection accuracy on the unlabeled target domain [9], [10], [11], [12], [13]. Most existing DA object detection methods are still based on convolutional neural network (CNN)-based two-stage detection frameworks, such as Faster R-CNN [14], and utilize carefully designed feature alignment modules to enforce domain-invariant feature extraction. By aligning features from both the source and target domains, these methods aim to enable the model to perform effective object detection tasks in the target domain, even in the absence of labeled data from the target domain. Among them, Chen et al. [15] first propose a DA Faster R-CNN framework that aligns both image-level and instance-level features to reduce domain discrepancy for object detection in the wild. Saito et al. [16] propose a strong-weak distribution alignment framework for DA object detection, which utilizes both local strong alignment and global weak alignment to bridge the domain gap. Zhou et al. [17] propose a multigranularity alignment method for DA object detection, achieving feature alignment across pixel, instance, and category levels. Guo et al. [18] propose a dual semantic correlation alignment method for DA object detection, focusing on aligning semantic information between source and target domains through context and class correlation.

Although the aforementioned DA object detection methods have achieved significant success in natural images, their performance often drops considerably when applied to RSIs. This is due to the significant differences between remote sensing and natural images, including variations in object scale, shape, texture, and background complexity [19], [20], [21]. Furthermore, objects in RSIs tend to be more sparsely distributed and are often accompanied by substantial noise and occlusions [22], [23], making it challenging for traditional DA methods to effectively address these issues. To alleviate these problems, some scholars have recently begun to explore DA object detection strategies based on the characteristics of RSIs.

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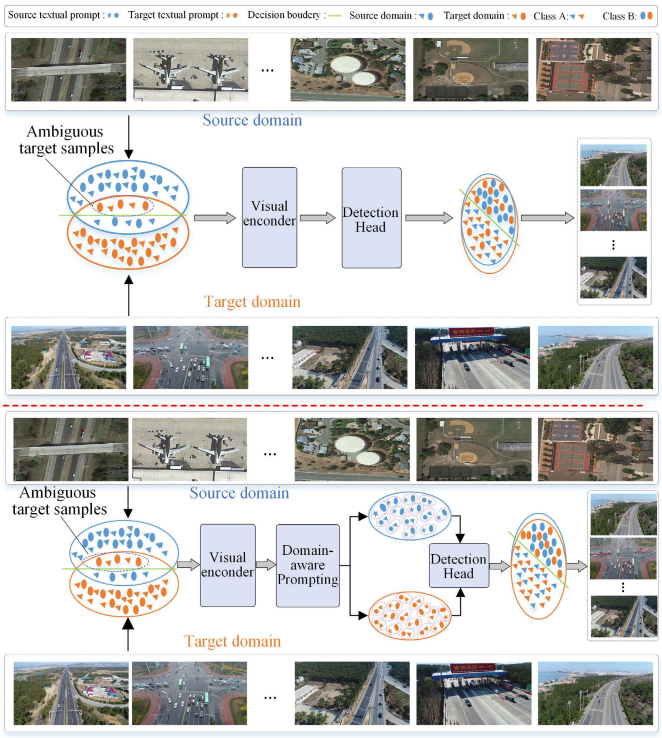


Fig. 1. Motivation of the proposed method. Top: Existing methods typically aim to reduce domain bias in object detection by optimizing discriminative visual encoders within the detection backbone. However, such strategies may overlook domain-aware semantic consistency, resulting in misalignment between features from the source and target domains. Bottom: our method learns domain-invariant semantics by modeling the learnable domain-aware prompts.

Han et al. [6] propose a DA detection transformer method, which effectively improves DA object detection accuracy in RSIs by combining transformer architecture, frequency-enhanced feature alignment, and a learnable alignment module. Luo et al. [7] present a self-training-based DA method for object detection in RSIs. Zhu et al. [24] propose a reconstructed feature alignment network, which enhances DA in object detection for RSIs by reconstructing and aligning features across domains to improve detection accuracy. Biswas and Tešić [25] propose a contrastive learning-based approach to address the DA issue in object detection tasks for satellite images. Although the above methods have achieved great success, they mainly emphasize mitigating the domain bias of the detection backbone by inferring discriminative visual encoders. As illustrated in Fig. 1, these coarse alignment approaches between the two domains may disrupt semantic feature consistency, causing misalignment of features. The misalignment can adversely affect the accuracy of detection in the target domain, especially for the remote sensing scene with sparse objects and diverse backgrounds.

Recently, visual language models (VLMs) have demonstrated remarkable generalization abilities. By leveraging prompt-based learning, VLMs utilize knowledge gained from large-scale image–text pair pretraining to perform logical reasoning and transfer prior knowledge, ultimately strengthening their capacity to solve unseen problems. It is noteworthy that the pioneering VLM contrastive language-image pretraining (CLIP) utilizes class labels as textual prompts during its training process. This prompt-based mechanism enables the construction of semantically richer representations within a unified embedding space. Such as, Ge et al. [26] introduced a DA approach based on prompt learning for image classification. Singha et al. [27] introduced a domain-agnostic prompt learning approach for CLIP, aiming to tackle the DA challenge in image classification. Li et al. [28] proposed a domain-aware detection head with prompt tuning for the object detection task. These prompt-based methods have achieved significant success in natural images. Similarly, the prompt-based methods play a crucial role in RSIs, as remote sensing data are often influenced by various factors such as different sensors, geographic environments, and sampling conditions, resulting in notable distribution differences between the source and target domains. Prompt-based learning helps establish cross-domain associations in object detection by providing specific semantic descriptions for the objects in RSIs. By combining image features with textual descriptions, meaningful contextual information can be transferred across domains, reducing the semantic gap between images and text.

Additionally, in the context of RSIs, CLIP can more effectively represent remote sensing scenes and objects through prompt-guided alignment between textual labels and visual features. This approach not only enhances the model's performance across various geographical environments but also improves its adaptability to different sensor modalities, thereby significantly boosting the model's generalization capabilities. Liu et al. [29] proposed the RemoteCLIP framework, a vision-language model tailored for remote sensing, designed to learn semantically rich visual features and align them with text embeddings for downstream applications. Song et al. [30] introduced a rotation-robust CLIP model that enhances the classification of RSIs by incorporating a visual adapter and a dual text prompt mechanism for improved rotation resilience. Zhang et al. [31] introduced a prompting strategy utilizing linguistic descriptions to facilitate CLIP in generating semantically varied contextual information for RSIs. Although most existing prompt learning methods optimize prompts in both the text and visual domains simultaneously, they fail to fully account for the reciprocal learning between text and visual prompts and are unable to flexibly adjust text and visual representations to better align with downstream object detection tasks. When dealing with complex DA tasks, these straightforward CLIP methods may not yield the best results.

To overcome the above challenge, we propose a domain-aware prompting (DAP) framework. This framework fully leverages CLIP model knowledge [32] and source knowledge to learn domain-invariant aware features for DA object detection in RSIs, as shown in Fig. 2. The DAP consists of two modules, namely reciprocal prompt interaction learning (RPIL) and prototype-driven domain prompt alignment (PDPA) modules. The RPIL module learns domain-invariant features by reciprocally modeling visual prompts with textual semantics and text prompts with visual semantics. Specifically, we propose a reciprocal learning mechanism in which visual features are enhanced by text-guided prompt learning, while textual prompts are simultaneously enriched through

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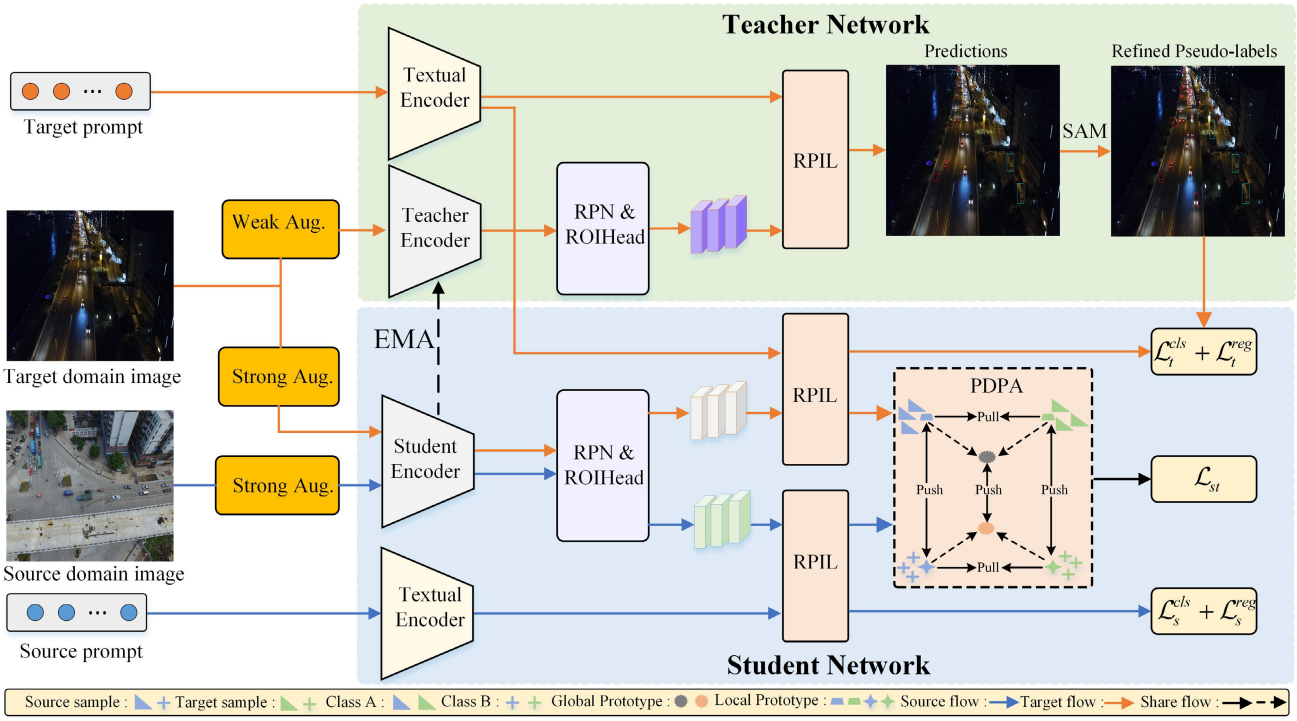


Fig. 2. Overall framework of our adaptation network. The RPIL module is deployed to generate domain-aware semantics for each domain by modeling reciprocal interactions between textual and image modalities. The PDPA module captures shared domain-aware prompt semantics from both local and global levels by constructing domain prompt prototypes.

visual-guided learning. The reciprocal enables dynamic alignment between visual and textual modalities, encouraging the extraction of domain-invariant semantics. By continuously refining this text–image alignment, the framework effectively bridges the semantic gap and facilitates the learning of domain-invariant representations.

To facilitate the alignment of the learnable domain prompts, we propose a PDPA module. This module is designed to learn domain-shared prompt prototypes that serve as a bridge for transferring shared categorical knowledge. Specifically, since the pseudo-labels generated from the teacher network can be noisy, we first utilize the segment anything model (SAM) [33] to produce accurate class-agnostic mask predictions to refine and improve the quality of these pseudo-labels. Subsequently, we incorporate domain-specific category information to further model the domain prompts into local and global prototypes, enabling the capture of both local and global semantic representations for more effective cross-domain alignment.

The integration of RPIL and PDPA results in a novel DAP framework that supports end-to-end training and enables effective adaptation for DA object detection in RSIs. The main contributions of our proposed approach are outlined as follows.

1) We propose a novel DAP framework for DA object detection in RSIs. By leveraging CLIP, our framework learns text prompts to effectively transfer both pretrained and source-domain knowledge to the target domain.
2) We design the RPIL and PDPA modules. The RPIL module employs reciprocal learning between textual and visual prompts to encourage the extraction of domain-invariant features across modalities. The PDPA module achieves domain prompt alignment from both local and global levels by modeling domain prompt prototypes.
3) Extensive experimental results show that the proposed approach surpasses existing advanced methods and proves to be broadly applicable in diverse cross-domain scenarios.

II. Related Works

A. Domain Adaptive Object Detection

DA employs labeled data from the source domain to facilitate model training, thereby enabling adaptation to the target domain and yielding enhanced performance without the necessity of target-domain annotations [34], [35], [36], [37]. Li et al. [38] proposed a semantic-complete graph-matching framework for DA object detection, which addresses mismatched semantics and facilitates DA by reformulating the problem as a graph-matching task. Cao et al. [39] introduced a contrastive mean teacher framework, which incorporates object-level contrastive learning to strengthen feature representations for DA object detection. Vibashan et al. [40] proposed a category-aware feature alignment method. By introducing category-aware discriminators, this approach effectively incorporates category information into the domain alignment process, significantly enhancing the model's adaptability. Zhao and Wang [41] proposed a method that reduces domain shift by aligning task-specific inconsistencies between classification and localization branches, improving cross-domain detection performance. Li et al. [42] proposed a teacher–student framework that innovatively combines domain adversarial learning

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with weak-strong data augmentation strategies for DA object detection.

The aforementioned methods offer feasible solutions for transferring knowledge from the source to the target domain in natural images for DA object detection. More recently, DA object detection has gained increasing attention in the field of RSIs. Building on established practices, Zhu et al. [43] introduced a dual-head rectified DA network. By combining coarse-to-fine consistency mechanisms with dual-head co-training, the network progressively adapts the data distributions. Ma et al. [44] introduced a hierarchical alignment network, which bridges the domain gap through the integration of image-wise and instance-wise alignment strategies, resulting in enhanced detection capabilities. Zhang et al. [45] proposed a foreground instance enhancement network, which alleviates the feature discrepancy between foreground and background and enhances the alignment of foreground features, thus enhancing the effectiveness of DA object detection in RSIs. Differently, we propose a novel detector based on a joint CLIP model, specifically designed for DA object detection in RSIs. This detector effectively addresses the challenge of feature alignment in complex scenes by introducing a reciprocal learning mechanism between text and visual features, especially suitable for remote sensing scenarios with sparse small objects and diverse backgrounds.

B. Prompt Learning in Vision-Language Models

CLIP is a prominent representative of VLMs, which is designed to train visual and text encoders through contrastive learning on a large-scale image–text pair dataset. CLIP consists of a visual encoder $\mathcal{F}$ and a text encoder $\mathcal{G}$, which collaboratively align visual and textual representations. For a classification task with C categories, CLIP constructs a textual embedding space using class names combined with a handcrafted prompt. This process begins with a text tokenizer $e(\cdot)$, which converts text descriptions into vector representations, followed by a text encoder $\mathcal{G}(\cdot)$, which maps these vectors into the embedding space. Specifically, for a given class name class-c, a fixed prompt t_c is formulated using the template A photo of a [class-i]. The tokenizer processes this prompt as $t_c = e(\text{A photo of a [class-c]})$, and the text encoder subsequently maps it into the textual embedding space.

However, manually designed prompts can often be simplistic and suboptimal. A more effective approach, inspired by CoOp [46], is to make the prompt representations learnable. Specifically, instead of relying on fixed handcrafted templates, we optimize a set of L learnable tokens. Formally, the traditional prompt is replaced with a learnable parameter matrix: $u = [u_1][u_2]\cdots[u_L]$, where each u_i is a trainable token. The learnable prompt t_c is then formulated as a combination of the set of learnable tokens u and the tokenized vector $b_c = e(\text{class-c})$, which encodes the representation of the cth class. Therefore, the prompt t_c is expressed as follows:

$$t_c = [u_1][u_2]\cdots[u_L][b_c]. \tag{1}$$

The above-mentioned prompt-based learning methods have primarily been applied to natural image scenes. In recent years, prompt learning has garnered increasing attention in the field of RSIs. Rambabu et al. [47] proposed SuperCLIP, a semantic attribute-guided transformer model that combines super-resolution and CLIP for zero-shot remote sensing scene classification. Cao et al. [48] introduced the M^{3}amba framework, a CLIP-based Mamba model developed for multimodal fusion to address the remote sensing classification challenge. Dai et al. [49] introduced a prompt-based multimodal fusion framework, termed ProMF, designed for the classification of remote sensing multimodal images. However, these methods overlook cross-domain discrepancies, resulting in domain-agnostic prompts that only capture domain knowledge from a single training set. To enable learning cross-domain information, we propose a novel DA prompting approach.

III. Method

A. Overview of DA Object Detection

1) Problem Definition: In the DA object detection setting, we define the labeled source domain $\mathcal{D}_s : \{x_s^i, y_s^i\}_{i=1}^{\mathcal{N}_s}$ and unlabeled target domain $\mathcal{D}_t : \{x_t^i\}_{i=1}^{\mathcal{N}_t}$, respectively. $\mathcal{D}_s$ and $\mathcal{D}_t$ derived from the source distribution $\mathcal{P}_s$ and target distribution $\mathcal{P}_t$ ($\mathcal{P}_s \neq \mathcal{P}_t$), respectively. The DA object detection method is carried out using both $\mathcal{D}_s$ and $\mathcal{D}_t$. By leveraging this algorithm, the detector $\mathcal{M}$ is ultimately obtained and is capable of achieving robust performance in the target domain $\mathcal{D}_t$.

2) Framework: Fig. 2 presents our proposed framework, which aims to enhance performance in the target domain. The network integrates prompt learning with a teacher–student paradigm, in which the student model receives input solely from annotated source images. Typically, the teacher model generates pseudo labels for the unlabeled target domain, which are then used to guide the training of the student model. However, since the pseudo-labels generated by the teacher model can be noisy, they may hinder the optimal training of the student model, leading to degraded performance and weaker generalization. To address this challenge, we leverage SAM's robust class-agnostic mask predictions to refine pseudo proposals, ensuring higher quality supervision and improving the overall learning process. Regarding the detector's performance, unlabeled data undergoes weak augmentation before being fed into the teacher network, while strongly augmented versions are input into the student network. Moreover, the teacher model is progressively refined by updating its parameters with those of the student model via an exponential moving average (EMA), promoting stable and consistent learning. To alleviate domain shift in DA object detection, a novel DAP approach combining RPIL and PDPA is introduced. The RPIL captures domain-invariant semantics by reciprocally learning between visual and textual embeddings. The PDPA dynamically models a prototype space using class prototype prompts, facilitating the extraction of shared prototype knowledge. This approach enables the model to effectively learn domain-invariant features, bridging the gap between the source and target domains.

B. Reciprocal Prompt Interaction Learning

Recently, a variety of CLIP-based detectors have been introduced to efficiently leverage textual knowledge for object

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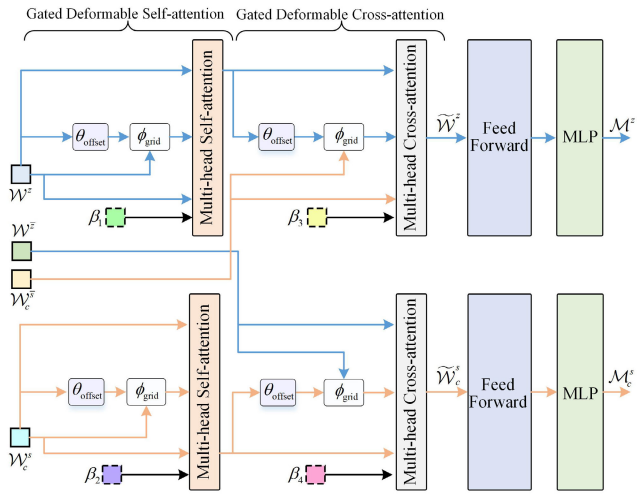


Fig. 3. RPIL module. The RPIL module facilitates the extraction of domain-invariant semantics by modeling reciprocal interactions between textual and visual prompts.

recognition. For example, DetPro [50] employs learnable prompts to produce domain-agnostic representations, further enhancing model adaptability across diverse scenarios. However, learning domain-agnostic prompts in DA, as described in (1), poses significant challenges. First, since the visual encoder is pretrained without explicit DA constraints, the resulting visual embeddings tend to exhibit domain bias and follow distinct distributions across different domains. Second, the high variability among instances leads to substantial intraclass divergence, making it difficult to align all samples with a unified class-level textual prompt. To address these challenges, we design an RPIL module that reciprocally learns visual prompts and textual prompts by modeling a gated deformable attention to promote more domain-agnostic visual representations, as shown in Fig. 3. The PRIL module consists of gated deformable self-attention (GDSA) and gated deformable cross-attention (GDCA). Specifically, GDSA enhances the discriminative power of visual or textual prompts by dynamically adjusting the focus area of the GDSA mechanism, allowing the model to more flexibly focus on key features within the object regions of the domain. Meanwhile, GDCA strengthens the category information in visual semantics by dynamically adjusting the focus area of the gated deformable cross-modal attention mechanism, thereby promoting the collaborative interaction between visual and textual prompts and facilitating the efficient fusion of cross-modal features. Here, given an input image x, it is first processed by a visual encoder $\mathcal{F}$, followed by a region proposal network (RPN) to generate a fixed-size region ofinterest (RoI) $\mathcal{W}^{\bar{z}} \in \mathbb{R}^{\widetilde{N}\times\widetilde{H}\times\widetilde{W}\times\widetilde{D}}$, where $\widetilde{N}$, $\widetilde{H}$, $\widetilde{W}$, and $\widetilde{D}$ are the batch size, height, width, and the channel number. We then use a global pooling operation to transform $\mathcal{W}^{\bar{z}}$ into the final visual embedding, i.e., $\mathcal{W}^{z} \in \mathbb{R}^{\widetilde{N}\times\widetilde{D}}$. For the learning prompt t_c, we input it into the text encoder to obtain text embedding features $\mathcal{W}_c^s$ and text context features $\mathcal{W}_c^{\bar{s}}$, i.e., $\mathcal{W}_c^s, \mathcal{W}_c^{\bar{s}} = \mathcal{G}(t_c)$.

To significantly enhance the discriminative power of both visual and textual embedding features, we first propose a new GDSA mechanism. By seamlessly integrating the dynamic sampling capability of deformable attention with the information-filtering mechanism of gated control, our method effectively improves the model's ability to capture fine-grained objects. Specifically, when the query ($\mathcal{Q}_1$), key ($\mathcal{K}_1$), and value ($\mathcal{V}_1$) are all set to the same visual embedding feature $\mathcal{W}^z$, we use the learnable offset parameters to adjust the sampling positions of the Key dynamically. This strategy allows the attention mechanism to adaptively focus on more informative regions rather than uniformly attending to all features. In particular, each Query is assigned a dedicated offset prediction function (offset projection) to compute its corresponding visual offsets

$$\Delta\mathcal{Q}_1 = \theta_{\text{offset}}(\mathcal{Q}_1) \tag{2}$$

where $\theta_{\text{offset}}(\cdot)$ is a learnable fully connected layer. When we take the text embedding $\mathcal{W}_k^s$ as the input key ($\mathcal{K}_2$), query ($\mathcal{Q}_2$), and value ($\mathcal{V}_2$). Here, we utilize the key $\mathcal{Q}_2$ to calculate the corresponding text offset

$$\Delta\mathcal{Q}_2 = \theta_{\text{offset}}(\mathcal{Q}_2). \tag{3}$$

These offsets are used to guide the sampling process of the key, ensuring that the model flexibly captures the most critical features of the visual and text. The calculation is as follows:

$$\mathcal{K}_1' = \phi_{\text{grid}}(\mathcal{K}_1, \Delta\mathcal{Q}_1) \tag{4}$$

$$\mathcal{K}_2' = \phi_{\text{grid}}(\mathcal{K}_2, \Delta\mathcal{Q}_2) \tag{5}$$

where $\phi_{\text{grid}}(\cdot)$ represents the grid sample function. This method enables flexible sampling across different Key positions, improving the model's adaptability to deformed objects.

To model the gated mechanism, we introduce two learnable gated parameters β_1 and β_2 to dynamically adjust the final visual and text attention information, respectively,

$$\mathcal{O}_1 = \beta_1 \cdot \Phi_{\text{self}}(\mathcal{Q}_1, \mathcal{K}_1', \mathcal{V}_1) \tag{6}$$

$$\mathcal{O}_2 = \beta_2 \cdot \Phi_{\text{self}}(\mathcal{Q}_2, \mathcal{K}_2', \mathcal{V}_2) \tag{7}$$

where β_1 and β_2 are two distinct learnable parameters optimized via backpropagation, and $\Phi_{\text{self}}(\cdot)$ represents multihead self-attention operation.

Since t_c contains rich category semantic knowledge that is not affected by domain difference, we can use this semantic information to guide the generation of visual cues, aiming to stimulate visual features with domain-invariant. To achieve this goal, we design a GDCA mechanism to integrate textual semantics into the visual representation learning. Here, the query ($\mathcal{Q}_3$) is $\mathcal{O}_1$, and key ($\mathcal{K}_3$) and value ($\mathcal{V}_3$) from text context feature $\mathcal{W}^{\bar{s}}$. Given the close mechanistic similarity between GDCA and GDSA operations, under the guidance of textual contextual information $\mathcal{W}_c^{\bar{s}}$. The visual embeddings with textual category semantic information are generated as follows:

$$\widetilde{\mathcal{W}}^{z} = \beta_3 \cdot \Phi_{\text{cross}}\left[\mathcal{Q}_3, \phi_{\text{grid}}(\mathcal{K}_3, \theta_{\text{offset}}(\mathcal{Q}_3)), \mathcal{V}_3\right] \tag{8}$$

where β_3 represents the learnable gated parameter, $\Phi_{\text{cross}}(\cdot)$ represents the multihead cross-attention operation.

To facilitate text embeddings in learning relevant knowledge from visual contextual information, we take $\mathcal{O}_2$ as the query

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($\mathcal{Q}_4$) and construct key–value pairs with the visual context feature $\mathcal{W}^{\bar{z}}$ as the key ($\mathcal{K}_4$) and value ($\mathcal{V}_4$). Driven by visual context information $\mathcal{W}^{\bar{z}}$, we use GDCA operations to facilitate the interactive learning of visual embeddings and textual information, thereby yielding text embeddings that incorporate visual semantic information. The calculation is as follows:

$$\widetilde{\mathcal{W}}_c^s = \beta_4 \cdot \Phi_{\text{cross}}\left[\mathcal{Q}_4, \phi_{\text{grid}}\left(\mathcal{K}_4, \theta_{\text{offset}}\left(\mathcal{Q}_4\right)\right), \mathcal{V}_4\right] \tag{9}$$

where β_4 represents the learnable gated parameter. The output gated deformable attention features $\widetilde{\mathcal{W}}^z$ are passed through the feed-forward layer from the transformer [51] and two-layer multilayer perception (MLP) with SwiGLU activation [52] to obtain the most significant aggregated embedding features $\mathcal{W}^z_{\text{agg}}$. Finally, we obtain the visual embedding $\mathcal{M}^z$ through a residual connection: $\mathcal{M}^z = \mathcal{W}^z + \eta_z \mathcal{W}^z_{\text{agg}}$, where η_z is a learnable parameter to control the scaling of the residual. By integrating textual semantics into the visual representation learning process, our model can learn more domain-invariant features, leading to improved performance.

Similarly, the output text embeddings $\widetilde{\mathcal{W}}_c^s$ are processed through the transformer's feed-forward layer, followed by a two-layer MLP with SwiGLU activation, to generate the most significant aggregated text embeddings $\mathcal{W}^s_{\text{agg}}$. The final text embedding is obtained by: $\mathcal{M}_c^s = \mathcal{W}_c^s + \eta_s \mathcal{W}^s_{\text{agg}}$, where η_s is a learnable parameter to control the scaling of the residual. In this way, text embeddings incorporate visual knowledge to enhance contextual awareness.

Building upon the aforementioned operations, we obtain the interactively learned text and visual embedding features. These two prompt branches reciprocally guide each other, ensuring synergistic collaboration. With the support of these two branches, the prediction probability of the detection head is calculated

$$P\left(y = c \mid x\right) = \frac{\exp\left(\cos\left(\mathcal{M}^z, \mathcal{M}_c^s\right)/\tau\right)}{\sum_{c=1}^{C} \exp\left(\cos\left(\mathcal{M}^z, \mathcal{M}_c^s\right)/\tau\right)} \tag{10}$$

where $\cos(\cdot)$ represents the cosine similarity, τ is the temperature parameter. According to (10), we obtain the detection head prediction probabilities under different conditions: under the source prompt $\{t_c^s\}_{c=1}^C$, the prediction probability is denoted as P_s; under the target prompt $\{t_c^t\}_{c=1}^C$, the student network's detection head prediction probability is represented as P_t, and the teacher network's detection head prediction probability is denoted as P_{te}. These probability distributions reflect the detection results of different models under varying prompt conditions, providing a crucial foundation for further optimizing pseudo-labels of the target domain.

C. SAM-Guided Pseudo-Label Refinement

In DA object detection, the quality of pseudo-labels generated by the teacher network is crucial, as it directly impacts the learning effectiveness of the student model. To address the issues of high noise and low localization accuracy in traditional pseudo-labeling methods, we propose a SAM-guided pseudo-label refinement strategy. Specifically, we first select the top-k highest-confidence prediction boxes from the teacher network's output $\mathcal{H}_t^T$ as the initial pseudo-label candidates, ensuring high reliability. To account for slight object shifts, these candidate boxes are modestly enlarged to ensure better coverage of the object regions. Next, the expanded bounding boxes are transformed into a format compatible with SAM, which leverages its powerful segmentation capabilities to generate high-resolution object masks. From these masks, we extract more precise bounding boxes, thereby refining the localization of pseudo-labels. To ensure the robustness of this strategy, we introduce a fallback mechanism: if SAM produces invalid bounding boxes (e.g., containing infinite values), the system automatically reverts to the initial pseudo-labels, preventing unreliable information from interfering with training. Finally, the refined pseudo-labels $\mathcal{B}_t$ (bounding boxes, class labels, and confidence scores) are used to supervise the student model, significantly improving its adaptation across different domains. By incorporating SAM-guided refinement, we effectively reduce pseudo-label noise, enhance localization accuracy, and lead to improved detection performance.

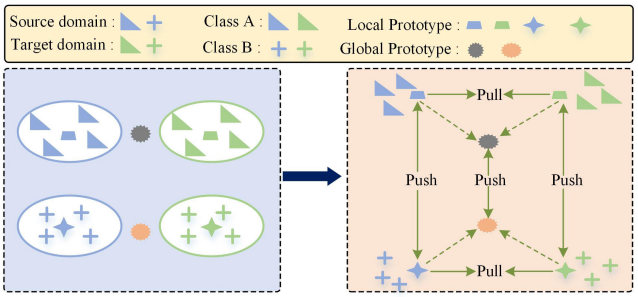


Fig. 4. PDPA module. The PDPA module performs domain prompt alignment by learning domain prompt local and global prototypes, enabling more effective domain-aware semantic consistency.

D. Prototype-Driven Domain Prompt Alignment

Since prototypes can effectively capture category-level feature representations and serve as stable class centers to guide domain feature alignment. Based on this idea, we propose a PDPA module to achieve cross-domain visual prompt alignment. To extract visual prompt prototypes, we introduce a prototype network at the output of the visual prompt in both the source and target domains. This prototype network maps the visual prompt into a d-dimensional feature space, capturing shared class representations across domains. As shown in Fig. 4, we construct a local prototype for each category in both the source and target domains (e.g., class A and class B in Fig. 4). These prototypes effectively capture domain-invariant prompt information. To capture the combined features of each category in both the source and target domains, we construct the global prototypes using the local prototypes of each class. We design a contrastive loss for the global prototypes, effectively separating different categories in the feature space and enhancing the discriminability between categories. At the same time, this loss aligns the local prototypes of each category in both the source and target domains with the global prototype, reducing confusion caused by appearance variations.

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In this article, we define the local prototypes in the source and target domains as $\mathcal{R}_s$ and $\mathcal{R}_t$, respectively, and their calculation is as follows:

$$\mathcal{R}_s = \mathcal{U}\left(\mathcal{M}_s^z\right), \quad \mathcal{R}_t = \mathcal{U}\left(\mathcal{M}_t^z\right) \tag{11}$$

where $\mathcal{U}(\cdot)$ represents a prototype network composed of a two-layer MLP, the term $\mathcal{M}_s^z$ represents the visual prompt feature of the student model in the source domain, and $\mathcal{M}_t^z$ corresponds to the visual prompt feature of the student model in the target domain. Here, we define the number of occurrences of class c in the source domain as b_s^c and the number of occurrences in the target domain as b_t^c. Due to the absence of labeled information in the target domain, we define the class pseudo-labels generated by the teacher network and filtered through SAM as p_t. In the source domain, the local prototype $\mathcal{R}_s$ is updated according to the class label p_s and the number of occurrences b_s^c of class c. In the target domain, the local prototype $\mathcal{R}_t$ is updated according to the pseudo-label p_t and the number of occurrences b_t^c of class c. The updated local prompt prototypes for the source and target domains are denoted as $\widetilde{\mathcal{R}}_s^c$ and $\widetilde{\mathcal{R}}_t^c$, respectively, and are computed as follows:

$$\widetilde{\mathcal{R}}_s^c = \widetilde{\mathcal{R}}_s^c \times \frac{b_s^c}{b_s^c+1} + \mathcal{R}_s \times \left(1 - \frac{b_s^c}{b_s^c+1}\right) \tag{12}$$

$$\widetilde{\mathcal{R}}_t^c = \widetilde{\mathcal{R}}_t^c \times \frac{b_t^c}{b_t^c+1} + \mathcal{R}_t \times \left(1 - \frac{b_t^c}{b_t^c+1}\right). \tag{13}$$

Here, when $b_s^c = 0$ and $b_t^c = 0$, the corresponding local prototype is not updated. As training progresses, b_s^c and b_t^c gradually increase. To achieve prompt-based cross-domain local prototype alignment, we introduce an explicit constraint. This constraint enforces consistent cross-domain representations, effectively reducing the distance between source-domain local prototypes $\widetilde{\mathcal{R}}_s$ and target-domain local prototypes $\widetilde{\mathcal{R}}_s$. Specifically, we construct a prompt-based local prototype alignment (PLPA) loss that leverages the contrastive InfoNCE loss [53] to pull prototypes of the same class closer together, while pushing prototypes from different classes farther apart

$$\mathcal{L}_{\text{PLPA}} = \frac{1}{C}\sum_{c=1}^{C}\left(-\log\frac{\exp\left(\left(\widetilde{\mathcal{R}}_s^c\right)^\top \widetilde{\mathcal{R}}_t^c\right)}{\sum_{c'}\exp\left(\left(\widetilde{\mathcal{R}}_s^c\right)^\top \widetilde{\mathcal{R}}_t^{c'}\right)}\right) \tag{14}$$

where $\top$ indicates the transpose operation.

To further push the prototypes of the different classes apart, we construct the global prototype. Specifically, we first assign weights to the local prototypes of class c based on its occurrence count in each domain. Then, we obtain the global prototype that more accurately reflects the class distribution by averaging these weighted local prototypes. The calculation is as follows:

$$\mathcal{Z}_s^c = \mathcal{Z}_s^c \times \frac{b_s^c}{\eta_s} + \widetilde{\mathcal{R}}_s^c \times \left(1 - \frac{b_s^c}{\eta_s}\right) \tag{15}$$

$$\mathcal{Z}_t^c = \mathcal{Z}_t^c \times \frac{b_t^c}{\eta_t} + \widetilde{\mathcal{R}}_t^c \times \left(1 - \frac{b_t^c}{\eta_t}\right) \tag{16}$$

$$\mathcal{Z}_{\text{pro}}^c = \frac{\mathcal{Z}_s^c + \mathcal{Z}_t^c}{2} \tag{17}$$

where $\mathcal{Z}_s^c$ and $\mathcal{Z}_t^c$ denote the local prototypes updated in the source and target domains, respectively, according to the occurrence frequency of class c. $\mathcal{Z}_{\text{pro}}^c$ represents the global prompt prototype obtained for class c. η_s represents the cardinality of the source-domain label p_s, η_t represents the cardinality of the target-domain label p_t. According to the $\mathcal{Z}_{\text{pro}}^c$, we design a prompt-based global prototype alignment (PGPA) loss that leverages the contrastive InfoNCE loss [53] to push prototypes from different classes farther apart

$$\mathcal{L}_{\text{PGPA}} = \frac{1}{C}\sum_{c=1}^{C}\sum_{c\neq i}^{C}\left(-\log\frac{\exp\left(\left(\mathcal{Z}_{\text{pro}}^c\right)^\top \mathcal{Z}_{\text{pro}}^i\right)}{\sum_{c'}\exp\left(\left(\mathcal{Z}_{\text{pro}}^c\right)^\top \mathcal{Z}_{\text{pro}}^{c'}\right)}\right). \tag{18}$$

The prompt-based alignment loss on prototypes can be expressed as

$$\mathcal{L}_{\text{PPA}} = \mathcal{L}_{\text{PLPA}} - \mathcal{L}_{\text{PGPA}}. \tag{19}$$

In addition to domain visual prompt alignment, the consistency between the source-domain and target-domain text prompts is also crucial for the model to learn domain-invariant features. To achieve this, we design a text prompt consistency (TPC) loss that minimizes the distance between the source-domain and target-domain text prompts. The loss function is defined as follows:

$$\mathcal{L}_{\text{TPC}} = \frac{1}{N_{\text{text}}}\sum_{m=1}^{N_{\text{text}}}\left(\mathcal{M}_c^{s,m} - \mathcal{M}_c^{t,m}\right)^2 \tag{20}$$

where N_{text} represents the number of text samples, $\mathcal{M}_c^{s,m}$ represents the updated source text embedding, and $\mathcal{M}_c^{t,m}$ represents the updated target text embedding. The final alignment loss on prompts can be expressed as

$$\mathcal{L}_{\text{align}} = \mathcal{L}_{\text{PPA}} + \mathcal{L}_{\text{TPC}}. \tag{21}$$

E. Model Optimization

As shown in Fig. 2, our network framework is optimized with three loss functions: the supervised detection loss $\mathcal{L}_{\text{det}}^{\text{sup}}$, the unsupervised detection loss $\mathcal{L}_{\text{det}}^{\text{unsup}}$, and the prompt-based domain alignment loss $\mathcal{L}_{\text{align}}$. Therefore, the training objective of the framework is defined as follows:

$$\begin{aligned}\mathcal{L}_{\text{all}} &= \mathcal{L}_{\text{det}}^{\text{sup}} + \lambda_{\text{unsup}}\mathcal{L}_{\text{det}}^{\text{unsup}} + \lambda_{\text{align}}\mathcal{L}_{\text{align}} \\ &= \left(\mathcal{L}_s^{\text{cls}} + \mathcal{L}_s^{\text{reg}}\right) + \lambda_{\text{unsup}}\left(\mathcal{L}_t^{\text{cls}} + \mathcal{L}_t^{\text{reg}}\right) + \lambda_{\text{align}}\mathcal{L}_{\text{align}} \\ &= \mathcal{L}_{\text{det}}\left(P_s, y_s\right) + \lambda_{\text{unsup}}\mathcal{L}_{\text{det}}\left(P_t, \mathcal{B}_t\right) + \lambda_{\text{align}}\mathcal{L}_{\text{align}}\end{aligned} \tag{22}$$

where $\mathcal{L}_{\text{det}}$ is the detection loss of Faster-rcnn. λ_{unsup} and λ_{align} are hyper-parameters that balance the loss terms.

IV. Experiments

A. Datasets and Experimental Setup

We conduct extensive evaluations on five publicly available remote sensing datasets to validate the effectiveness and generalization ability of the proposed method. The public datasets include VisDrone [59], UAVDT [60], DIOR [61], SSDD [62], and HRSID [63].

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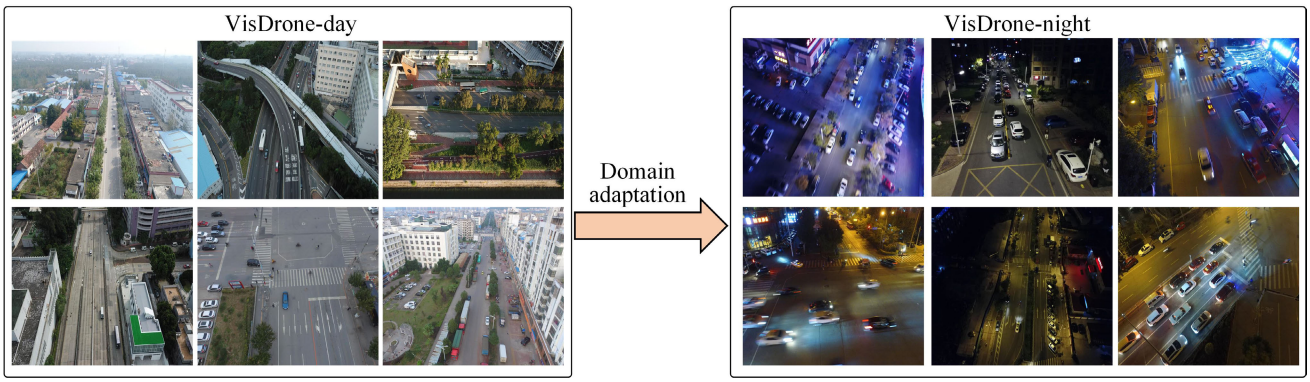


Fig. 5. Some examples are taken from the Visdrone-day and Visdrone-night datasets, respectively. We construct a cross-condition adaptation for knowledge transfer from the VisDrone-day (source domain) to the VisDrone-night (target domain) datasets.

1) VisDrone: The public VisDrone dataset is a widely used aerial RSIs dataset comprising ten object categories. The images were captured by various uncrewed aerial vehicle (UAV) platforms under diverse weather and lighting conditions. The dataset contains a total of 8599 images, which are divided into 6471 for training, 1580 for validation, and 548 for testing.

2) UAVDT: The public UAVDT dataset is a UAV-based dataset comprising 80 000 image frames extracted from over ten hours of video footage captured in various urban environments. The videos were recorded using UAV platforms at different altitudes and viewpoints to reflect diverse aerial perspectives. The dataset is split into 5000 training images, 1500 validation images, and 3500 testing images, each with a resolution of 1024×540 pixels. UAVDT focuses on three object categories: car, bus, and truck.

3) DIOR: The public DIOR dataset is a large-scale satellite dataset comprising 23 463 images with annotations for 190 288 object instances across 20 categories. The dataset is divided into 16 424 training images, 2346 validation images, and 4693 testing images, each with a fixed resolution of 800×800 pixels.

4) SSDD: The public SSDD dataset is a publicly available benchmark released by the Naval University of Engineering, China, for ship detection in SAR imagery. It comprises 1160 high-resolution SAR images with a spatial resolution of 1 meter and includes 2456 annotated ship instances. The images were collected from various geographic regions under diverse environmental conditions, such as open seas, nearshore waters, and land-sea transitional zones.

5) HRSID: The public HRSID dataset is a publicly available benchmark dataset for ship detection in high-resolution synthetic aperture radar (SAR) imagery. The dataset is divided into 3642 training images and 1962 testing images, each with a fixed resolution of 800×800 pixels. It is widely used to evaluate the robustness and generalization capability of object detection algorithms in SAR-based remote sensing scenarios.

To rigorously evaluate the effectiveness and generalization capacity of the proposed DAP framework, we design comprehensive experiments under three representative and challenging DA scenarios, namely: 1) cross-condition adaptation, 2) cross-view adaptation, and 3) cross-modality adaptation. These scenarios collectively cover various shifts in the data distribution, thus providing a thorough assessment of the adaptability of DAP. Table I provides a detailed description of these adaptation scenario datasets. Cross-condition adaptation focuses on variations caused by imaging conditions while maintaining consistent viewpoints and sensor types. In this scenario, all datasets are collected using UAV sensors, ensuring that domain shifts arise solely from changes in imaging conditions, and we set up three sets of experiments.

1) From the VisDrone dataset, we select 6694 daytime images and 1823 nighttime images to construct two new subsets, named VisDrone-day and VisDrone-night, respectively. Some example scene images are shown in Fig. 5, where the left image serves as the source domain and the right image serves as the target domain in our experiments. Due to the limited number of samples, certain object categories in the nighttime images (such as bicycle, tricycle, and awning-tricycle) are excluded from the experiments to ensure reliable training and evaluation.
2) We select 20 891 daytime images from the UAVDT dataset to form the source domain, denoted as UAVDT-day, and additionally choose 11 489 nighttime images to construct the target domain, denoted as UAVDT-night.
3) We select 6404 daytime images from the VisDrone dataset to construct a new source domain, denoted as VisDrone-day-1, and adopt UAVDT-night as the target domain. Both domains share three common object categories: bus, truck, and car. Cross-view adaptation aims to assess the impact of viewpoint changes on detection performance, which focuses on the significant differences that exist in viewing angles and object scales, making this scenario particularly challenging.

In this setting, the source domain comprises satellite imagery collected from Google Earth, GF, and JL-1, while the target domain is composed of images captured by UAV platforms. We set up two sets of experiments.

1) We select 6421 satellite images related to the car category from the DIOR dataset to form the source domain, named DIOR-car, and choose 5364 daytime images related to the car category from the VisDrone dataset to form the target domain, named VisDrone-car.

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TABLE I
SUMMARY OF DIFFERENT DA SCENARIO DATASETS

DA Scenarios	Dataset	Sensor	Class
Cross-condition	VisDrone-day	UAV	pedestrian, people, car, van, truck, bus, motor
	VisDrone-night	UAV	pedestrian, people, car, van, truck, bus, motor
	UAVDT-day	UAV	car, truck, bus
	UAVDT-night	UAV	car, truck, bus
	VisDrone-day-1	UAV	car, truck, bus
Cross-view	VisDrone-car	UAV	car
	DIOR-car	Google Earth, GF, JL-1	car
	UAVDT-car	UAV	car
Cross-modality	SSDD	RadarSat-2, Sentinel-1, TerraSAR-X	ship
	DIOR-ship	Google Earth, GF, JL-1	ship
	HRSID	Sentinel-1, GF-3	ship

2) DIOR-car as the source domain, we select 7217 daytime images related to the car category from the UAVDT dataset to form the target domain, named UAVDT-car. Cross-modality adaptation is designed to evaluate the effectiveness of the proposed method under heterogeneous sensor modalities.

In this setting, the source domain is composed of optical RSIs, while the target domain consists of SAR images, and we set up two sets of experiments.

1) We select 2702 satellite images related to the ship category from the DIOR dataset to form the source domain, named DIOR-ship, and the SSDD dataset as the target domain.
2) DIOR-ship as the source domain and the HRSID dataset as the target domain.

B. Implementation Details

We carry out the proposed method on the PyTorch [64] framework, and the GPU is an NVIDIA 3090 with 24 GB of memory. The training of the student network was conducted with an initial learning rate set to 0.001. The training process is divided into two stages: the first 50,000 iterations constitute the burn-in phase, which focuses on transferring knowledge from the source domain to the target domain; in the subsequent 50 000 iterations, corresponding to the DA object detection phase, we set the EMA decay rate to 0.9996 for updating the teacher model. For the backbone network, we utilize Faster-RCNN with pretrained ResNet-50 as the primary object detector. We employ the mean accuracy precision (mAP) as an evaluation metric to evaluate the performance of the proposed method and some advanced methods. For weighting factors of λ_{unsup} and λ_{align}, we fix $\lambda_{\text{unsup}} = 1.0$ and $\lambda_{\text{align}} = 1.0$ in our all experiments.

C. Experimental Results and Analyses

We compare our method with other advanced DA object detection methods: SWDA [16], SIGMA [54], CMT [39] + PT [55], CMT [39] + AT [42], Cooperative-Students [56], DPA [57], and DATR [58]. Based on the experimental settings of cross-condition adaptation, cross-view adaptation, and cross-modality adaptation, we construct the following experiments.

1) Cross-Condition Adaptation: Tables II–IV present the experimental results of VisDrone-day → VisDrone-night, UAVDT-day → UAVDT-night, and VisDrone-day-1 → UAVDT-night, respectively. The proposed DAP achieves superior performance over all comparison methods in DA tasks involving changes in lighting conditions. Specifically, as evidenced in Table II, compared with the existing predominant DA methods for object detection, DAP achieves significantly better performance, surpassing SWDA, SIGMA, CMT + PT, CMT + AT, Cooperative-Student, DPA, and DATR methods by 13.89% (32.67% versus 18.78%), 11.91% (32.67% versus 20.76%), 10.76% (32.67% versus 21.91%), 9.36% (32.67% versus 23.31%), 4.60% (32.67% versus 28.07%), 4.49% (32.67% versus 28.18%), and 2.02% (32.67% versus 30.65%) in mAP, respectively. Meanwhile, DAP achieves a 17.14% (32.67% versus 15.53%) improvement in mAP over the nonadaptive (Source-only) baseline.

Tables III and IV present the cross-domain detection results from UAVDT-day to UAVDT-night, and from VisDrone-day-1 to UAVDT-night, respectively. The DAP method achieves the highest mAP in both experiments (34.84% and 43.50%, respectively), demonstrating superior cross-domain adaptability and category discrimination capability. Although the CMT + PT method shows exceptionally high precision in detecting the car category (71.33% in Tables III and 68.43% in Table IV), its performance on the truck and bus categories is significantly poorer, leading to a lower overall mAP. This suggests that the model may have misclassified truck and bus instances as cars under nighttime conditions, resulting in an artificially inflated precision for the car category. In Table III, DPA achieves a detection accuracy of 68.53% in the car category, higher than DATR's 57.24%. However, DPA performs weaker in the truck and bus categories, especially in the truck (DPA: 5.31%, DATR: 3.31%). Despite DPA's better performance in the car category, DATR's mAP (32.87%) is slightly higher than DPA's 31.82%. In Table IV, DPA leads DATR in the car category with 69.96%, compared to DATR's 59.88%. However, in the bus category, DATR surpasses DPA with 50.39%, while DPA stands at 32.79%. Overall, DATR's mAP is 40.82%, higher than DPA's 38.56%. In contrast, DAP achieves notably higher accuracy in detecting the visually similar yet semantically different categories of truck and bus, indicating greater robustness in complex nighttime scenarios.

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TABLE II
DETECTION RESULTS (%) FROM VISDRONE-DAY TO VISDRONE-NIGHT

Method	pedestrian	people	car	van	truck	bus	motor	mAP
Source-only	24.40	0.14	47.42	15.62	2.02	14.36	4.75	15.53
SWDA [16]	26.25	0.22	50.63	19.71	5.84	24.87	3.97	18.78
SIGMA [54]	26.39	0.30	51.78	22.44	10.02	30.25	4.18	20.76
CMT [39]+PT [55]	24.27	0.37	49.65	25.45	13.44	35.87	4.28	21.91
CMT [39]+AT [42]	24.48	2.61	54.74	19.35	19.73	32.53	9.71	23.31
Cooperative-Student [56]	37.51	5.02	51.47	29.64	23.57	34.87	14.40	28.07
DPA [57]	38.43	4.46	46.68	32.12	23.43	32.15	20.04	28.18
DATR [58]	41.65	**8.38**	50.70	**32.56**	22.59	**36.23**	22.42	30.65
DAP (Ours)	**44.11**	7.11	**61.89**	27.69	**28.57**	35.79	**23.50**	**32.67**

TABLE III
DETECTION RESULTS (%) FROM UAVDT-DAY TO UAVDT-NIGHT

Method	car	truck	bus	mAP
Source-only	52.32	0.17	0.52	17.67
SWDA [16]	57.78	2.03	2.15	20.65
SIGMA [54]	58.46	9.45	8.65	25.52
CMT [39]+PT [55]	**71.33**	0.63	10.38	27.45
CMT [39]+AT [42]	64.84	9.27	11.71	28.60
Cooperative-Student [56]	67.04	6.12	22.06	31.74
DPA [57]	68.53	5.31	21.64	31.82
DATR [58]	57.24	3.31	22.06	32.87
DAP (Ours)	61.60	**16.95**	**25.98**	**34.84**

TABLE IV
DETECTION RESULTS (%) FROM VISDRONE-DAY-1 TO UAVDT-NIGHT

Method	car	truck	bus	mAP
Source-only	28.15	0.98	26.66	18.60
SWDA [16]	48.26	1.04	27.46	25.59
SIGMA [54]	63.53	1.15	28.13	28.27
CMT [39]+PT [55]	68.43	3.22	26.59	32.75
CMT [39]+AT [42]	**70.63**	3.89	31.56	35.36
Cooperative-Student [56]	68.41	7.32	38.05	37.73
DPA [57]	69.96	12.94	32.79	38.56
DATR [58]	59.88	12.20	50.39	40.82
DAP (Ours)	64.18	**15.81**	**50.51**	**43.50**

TABLE V
DETECTION RESULTS (%) ON DIOR-CAR → VISDRONE-CAR, DIOR-CAR → UAVDT-CAR, RESPECTIVELY

Setting	DIOR-car → VisDrone-car	DIOR-car → UAVDT-car
Method	*car* mAP	*car* mAP
Source-only	10.82	31.82
SWDA [16]	14.43	40.69
SIGMA [65]	16.15	43.39
CMT [39]+PT [55]	17.04	45.86
CMT [39]+AT [42]	19.92	47.13
Cooperative-Student [56]	20.40	49.49
DPA [57]	20.69	50.51
DATR [58]	22.79	53.63
DAP (Ours)	**25.76**	**55.57**

TABLE VI
DETECTION RESULTS (%) ON DIOR-SHIP → SSDD, DIOR-SHIP → HRSID, RESPECTIVELY

Setting	DIOR-ship → SSDD	DIOR-ship → HRSID
Method	*ship* mAP	*ship* mAP
Source-only	36.78	28.32
SWDA [16]	47.23	39.69
SIGMA [65]	55.79	47.26
CMT [39]+PT [55]	62.39	52.17
CMT [39]+AT [42]	64.34	54.92
Cooperative-Student [56]	68.31	55.82
DPA [57]	68.73	56.06
DATR [58]	69.77	57.67
DAP (Ours)	**71.66**	**59.94**

In summary, DAP not only improves overall object detection performance but also effectively mitigates category confusion caused by visual degradation, showing particularly strong results on hard-to-detect categories. The key factors contributing to this notable improvement are as follows.

1) The RPIL module is deployed to generate domain-invariant features by modeling reciprocal learning between textual and visual prompts for each domain.
2) The PDPA module extracts shared domain-aware semantic representations by building prompt prototypes that integrate both local and global contextual information.

2) Cross-View Adaptation: Table V shows the results of two cross-DA tasks: from DIOR-car to VisDrone-car, and from DIOR-car to UAVDT-car, respectively, reflecting the adaptability between object detection datasets under different views. This adaptation task is more complex due to domain differences caused by image view variations, compared to tasks involving only changes in sampling conditions like day-to-night adaptation. In this context, we argue that DAP explicitly learns semantically rich text prompts to promote reciprocal reasoning between textual and visual modalities, thus enhancing the category-level discriminability of visual features and reducing the negative impact of background noise. As shown in the table, the experimental results from the two transfer tasks, DIOR-car → VisDrone-car and DIOR-car → UAVDT-car, demonstrate that the model trained solely on the source domain (source-only) performs the worst on the target domain, indicating its limited generalization ability without DA. After applying DA methods (such as SWDA, SIGMA, CMT and its variants, Cooperative-Student, DPA, as well as DATR), the detection accuracy on the target domain improves significantly. Among all the methods, the proposed DAP achieves the best performance on both target-domain datasets, clearly outperforming existing approaches and demonstrating superior domain generalization and transfer adaptability.

3) Cross-Modality Adaptation: As shown in Table VI, DAP achieves the best detection performance across both DA tasks.

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TABLE VII
ABLATION STUDIES (%) ACROSS DIFFERENT MODULES ON THE VISDRONE-DAY → VISDRONE-NIGHT

Baseline	CLIP	SAM	RPIL	PDPA	mAP
✓					26.94
✓	✓				28.83
✓	✓	✓			29.31
✓	✓	✓	✓		31.07
✓	✓	✓	✓	✓	**32.67**

TABLE VIII
COMPARISON (%) OF DIFFERENT MODULES ON THE VISDRONE-DAY → VISDRONE-NIGHT

Standard attention	RPIL	Prototype-oriented	PDPA	mAP
				29.31
✓				30.09
	✓			31.07
		✓		29.98
			✓	31.02

Specifically, it attains mAP values of 71.66% and 59.94% for the DIOR-ship → SSDD and DIOR-ship → HRSID tasks, respectively, significantly outperforming all compared methods. These results demonstrate that the proposed approach possesses superior generalization capability and domain adaptability in cross-domain object detection scenarios. Compared to the traditional source-only setting, DAP achieves substantial performance gains, highlighting the limitations of models trained without any DA. Furthermore, when compared with mainstream DA methods such as SWDA, SIGMA, CMT+AT, CMT+PT, Cooperative-Student, DPA, and DATR. DAP consistently shows superior performance in both transfer settings. Notably, the improvement is more pronounced in the more challenging DIOR-ship → HRSID task. Overall, the strong performance of DAP verifies its effectiveness and advancement in cross-domain ship detection.

D. Model Analysis

In this section, we conduct a deeper analysis of the model by examining its ablation study and hyperparameter sensitivity.

1) Ablation Studies: To assess the effectiveness of our DAP, we conduct comprehensive ablation studies to evaluate the contributions of each component on the VisDrone-day → VisDrone-night. We reveal the contribution of CLIP by integrating it into the DA object detection framework. As shown in Table VII, the introduction of CLIP significantly improves the Baseline model from 26.94% to 28.83% in terms of mAP. In this work, the Baseline model adopts a teacher–student architecture for DA object detection, where the pseudo-labels generated by the teacher network are used directly without any parameterized constraints or filtering. This improvement is primarily attributed to the introduction of CLIP, which enables the transfer of rich semantic knowledge from pretraining to bridge the visual gap between the source and target domains.

Furthermore, we observe that incorporating SAM results in an effective performance gain for the model. The result highlights the importance of SAM in enhancing pseudo-label quality before supervision, with the observed performance gain demonstrating its effectiveness in overcoming the limitations of self-training. The introduction of RPIL significantly improves the Baseline + CLIP + SAM, increasing from 29.31% to 31.07% in terms of mAP. The performance gain is mainly due to the RPIL module's use of reciprocal learning between textual and visual prompts, which promotes the learning of domain-invariant representations for each domain. When the inclusion of the PDPA module leads to the best performance, achieving an mAP of 32.67%. This demonstrates the strong effectiveness of aligning both local and global prototypes across domains using prompt-based contrastive learning.

To further validate the effectiveness of the proposed RPIL and PDPA modules, we perform a comparative experiment on the VisDrone-day → VisDrone-night task by replacing each module with functionally comparable alternatives while keeping all other components and training settings fixed. We evaluated RPIL and PDPA by comparing RPIL with existing standard attention methods such as self-attention and cross-attention [51], and by comparing PDPA with prototype-oriented approaches [66]. As presented in Table VIII, RPIL improves the mAP from 30.09% obtained with standard attention mechanisms to 31.07%, yielding gains of + 0.98%. For PDPA, replacing it with a prototype-oriented approach results in an mAP of 29.98%, whereas adopting PDPA increases the score to 31.02%, achieving a + 1.04% improvement. These results further validate the effectiveness of the RPIL and PDPA modules.

2) Hyperparameter Sensitivities: We performed a sensitivity analysis on the balancing parameter between the unsupervised detection loss and the prompt-based domain alignment loss. The methodology employed in our study primarily involves two hyperparameters, namely λ_{unsup} and λ_{align}. We initially set λ_{align} to 1.0, as this value has been empirically determined to optimize the performance of our method. We then demonstrate the accuracy of VisDrone-day → VisDrone-night by varying λ_{unsup} from 0.0 to 1.2, as illustrated in Fig. 6(a). After fixing λ_{align}, we varied λ_{unsup} from 0.0 to 1.2, as shown in Fig. 6(b). The experimental results indicate that the optimal detection accuracy of 32.67% is achieved when $\lambda_{\text{unsup}} = 1.0$ and $\lambda_{\text{align}} = 1.0$.

Fig. 6(c) shows the detection results under different learning rates on the VisDrone-day → VisDrone-night task. The experimental results show that the model achieves the best detection accuracy of 32.67% when the learning rate is set to 0.001. Performance deteriorates when the learning rate is either too high (e.g., 0.01) or too low (e.g., 0.000001), with the lowest detection accuracy observed at a learning rate of 0.000001, which is 1.06%. These results indicate that the choice of learning rate is crucial for model training, as an appropriate learning rate can significantly improve model performance, while both high and low learning rates may hinder it.

E. Detection Visualization

In this section, to more clearly illustrate the differences in detection results among different methods, we select

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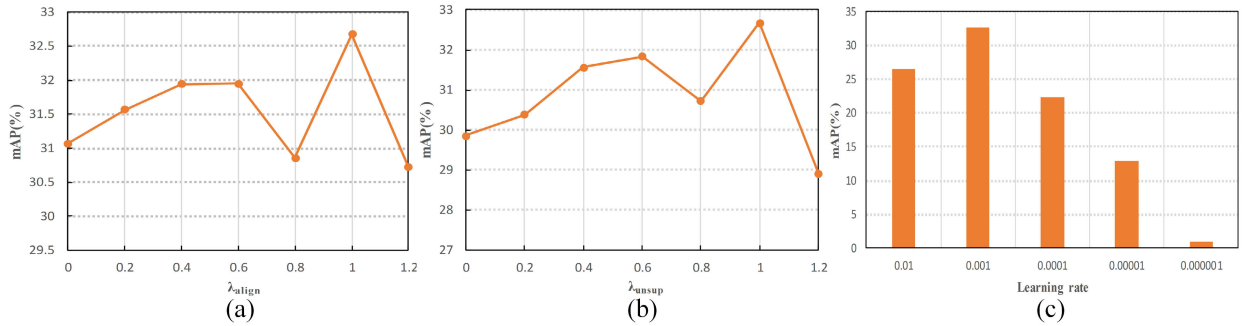


Fig. 6. Hyperparameter sensitivity analysis on the VisDrone-day → VisDrone-night. (a) Sensitivity analysis of parameter λ_{align}. (b) Sensitivity analysis of parameter λ_{unsup}. (c) Sensitivity analysis of learning rate.

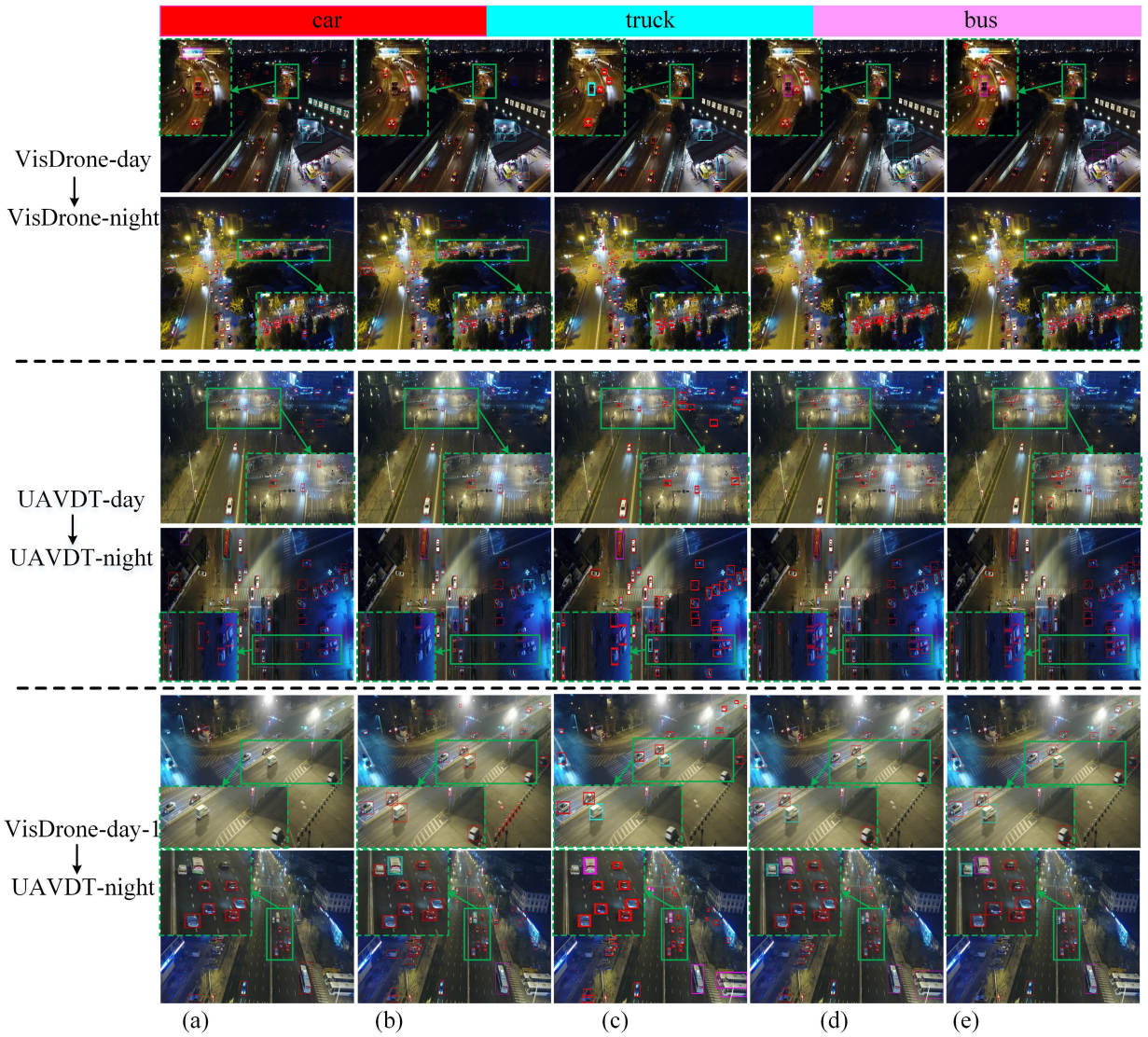


Fig. 7. Illustration of the detection results on cross-condition adaptation. (a)–(e) Represent the detection results of the source-only model, the CMT+AT, the DATR, the proposed DAP, and the ground-truth labels, respectively. Different categories are marked with different colors.

representative key regions from the outputs of each method (green solid boxes) and present enlarged views of these regions to highlight detailed information (green dashed boxes).

1) Cross-Condition Adaptation: Fig. 7 presents the results of DA experiments for object detection under three different time conditions from day to night. Each experiment includes detection outputs from "Source-only," "CMT + AT," "DATR,"

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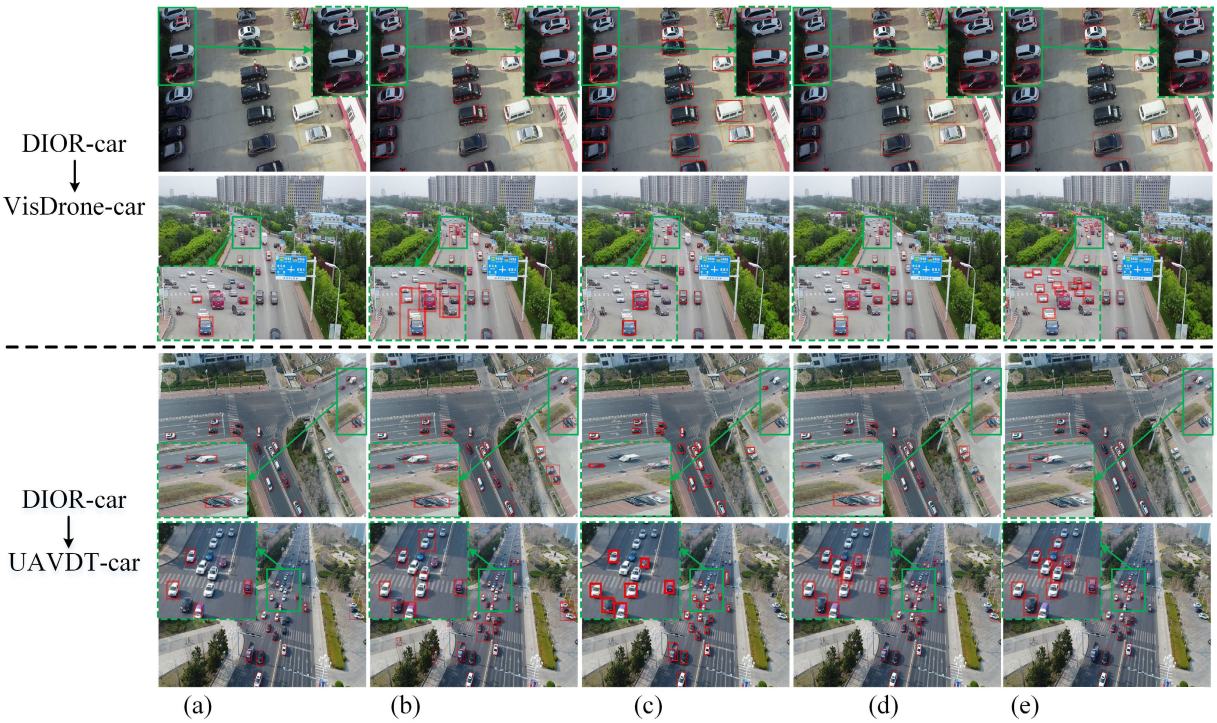


Fig. 8. Illustration of the detection results on cross-view adaptation. (a)–(e) Detection results of the source-only model, the CMT+AT, the DATR, the proposed DAP, and the ground-truth labels, respectively.

"Ours," and the "Ground truth." Overall, the results indicate that models trained solely on source-domain data ("Source-only") suffer from significant missed and false detections in nighttime scenes, with performance degradation particularly evident in regions with complex lighting. In contrast, the CMT + AT and DATR methods show improved detection accuracy, identifying more objects; however, they still struggle in challenging conditions such as small objects and occlusions. By comparison, the proposed method achieves more accurate and stable detection across all object categories (car, truck, and bus), and demonstrates notably better performance for difficult-to-identify objects such as the truck and bus categories. The detection results exhibit higher precision in bounding box localization and better completeness of object contours, aligning more closely with the ground truth. These findings validate the robustness and strong generalization ability of the proposed method for cross-domain nighttime object detection tasks.

2) Cross-View Adaptation: Fig. 8 shows the cross-domain object detection results from the DIOR-car dataset to the VisDrone-car dataset (top two rows) and the UAVDT-car dataset (bottom two rows). In the DIOR-car → VisDrone-car task, the source-only method demonstrates poor detection performance with a significant number of missed detections, indicating its limited ability to handle substantial domain shifts. Although the CMT + AT and DATR methods alleviate this issue to some extent, with improvements in both the number and positioning of detected objects compared to the source-only, they still exhibit omissions in detecting car class objects, showing noticeable discrepancies from the ground truth. In contrast, the proposed method achieves more accurate detection results in this task, with predicted bounding boxes more closely aligned with the ground truth, thereby improving detection accuracy. A similar pattern is observed in the DIOR-car → UAVDT-car task. The source-only method again suffers from poor performance due to domain shift, while CMT + AT and DATR provide moderate improvements yet struggle under complex backgrounds. The proposed method demonstrates superior adaptation capability on the UAVDT-car dataset, accurately detecting a greater number of car objects, further validating its effectiveness in DA object detection.

3) Cross-Modality Adaptation: Fig. 9 shows examples of detection when transferring from the DIOR-ship dataset to the SSDD and HRSID datasets, respectively, aiming to demonstrate the proposed method's adaptation performance across different target domains. In the DIOR-ship → SSDD and DIOR-ship → HRSID detection examples, our method demonstrates superior accuracy in detecting ship objects and significantly reduces false positives compared to other approaches. These results strongly indicate that our model retains excellent generalization capability, even in the presence of substantial differences in imaging mechanisms between the source domain (optical images) and the target domain (SAR images).

F. Model Complexity and Inference Efficiency

Table IX provides a comparative analysis between the proposed DAP and other algorithms with respect to model complexity [number of parameters and floating-point operations (FLOPs)] and inference efficiency (frames per second, FPS) under VisDrone-day → VisDrone-night adaptation setting. As illustrated in the table, SWDA is the most lightweight

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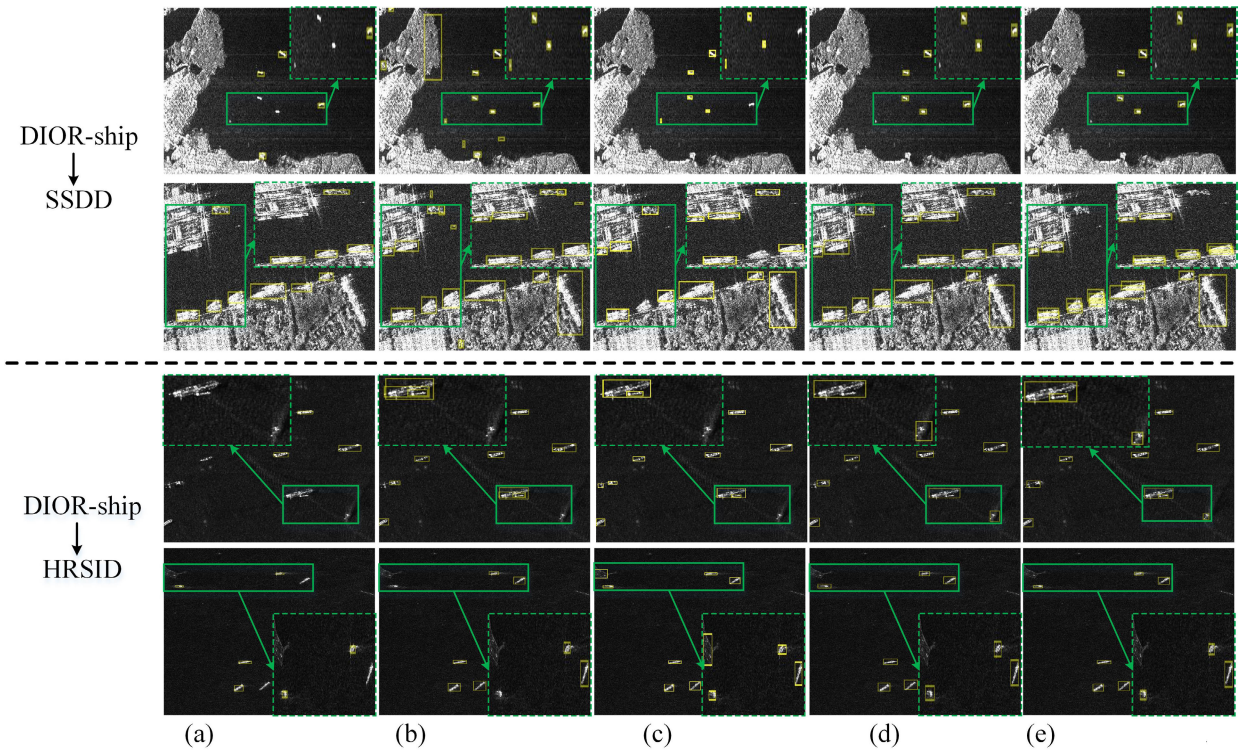


Fig. 9. Illustration of the detection results on cross-modality adaptation. (a)–(e) Detection results of the source-only model, the CMT+AT, the DATR, the proposed DAP, and the ground-truth labels, respectively.

TABLE IX
COMPARISON EXPERIMENTAL RESULTS OF THE PARAMETERS, FLOPS AND FPS UNDER VISDRONE-DAY → VISDRONE-NIGHT ADAPTATION SETTING

Methods	Parameters (M)	FLOPs(G)	FPS(f/s)
SWDA [16]	**33.84**	189.83	11.41
CMT [39]+PT [55]	70.50	275.32	19.10
CMT [39]+AT [42]	70.50	275.32	**19.50**
Cooperative-Student [56]	204.8	598.20	10.5
DPA [57]	47.37	**173.26**	18.42
DATR [58]	47.78	269.12	4.40
DAP (Ours)	307.20	2757.00	2.50

model, characterized by the smallest number of parameters (33.84 M). The DPA method stands out for its minimal computational complexity (173.26 G), paired with a comparatively high speed (18.42 f/s). Meanwhile, the CMT + AT achieves the fastest inference speed (19.50 f/s) among all evaluated models. In the proposed DAP method, the introduction of PRIL and SAM substantially improves the model's representational capability, while inevitably increasing the computational complexity. Consequently, the proposed approach exhibits the largest number of parameters (307.20 M) and FLOPs (2757.00 G) among all evaluated models.

V. CONCLUSION

In this article, we propose a novel DAP framework to address the issue of DA object detection in RSIs. Our framework involved two innovative modules: the RPIL module and the PDPA module. The RPIL module is designed to generate domain-aware semantics for each domain by modeling reciprocal interactions between textual and image modalities. The PDPA module is designed to capture shared domain prompt semantics at both local and global levels by constructing domain-invariant prompt prototypes. These two critical modules are closely integrated and mutually reinforcing, with their combination substantially enhancing domain prompt learning by approaching it from complementary perspectives. Extensive experiments validate the effectiveness of our method, demonstrating its superior performance over state-of-the-art approaches on different adaptation datasets in RSIs.

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